openSUSE immutable systems

# Harder to break. Easier to fix.

Immutable systems keep the core OS read-only and replace it as a whole unit. That changes what an attacker can make stick.

## WHAT IT HELPS WITH

- ✓ **Persistence is harder**
  Changes to /usr or boot files fail, or vanish on reboot.
- ✓ **Clean recovery**
  Roll back or redeploy to a pristine image.
- ✓ **Consistent patching**
  Atomic updates leave fewer half-updated machines.
- ✓ **Tampering stands out**
  A fixed known-good image makes drift visible.
- ✓ **Smaller attack surface**
  Minimal base; apps live in containers or Flatpaks.
- ✓ **Predictable base**
  Read-only roots and snapshots keep the OS stable.

## Built on openSUSE

**Leap**
Transactional server role

**MicroOS**
Read-only container host

**Kalpa**
Immutable KDE Plasma

Other downstream flavors from openSUSE may follow the same model.